

Binaural Audio-Visual Instance Segmentation

Saijun Wang^{a,1}, Guanfeng Tang^{a,1}, Hongbo Zhao^a, Zhicheng Lei^a, Yutong Zhang^a, Wei Ye^b and Rui Fan^{b,*}

^aCollege of Electronic and Information Engineering, Tongji University, Shanghai 201804, China,

^bCollege of Electronic and Information Engineering, Shanghai Institute of Intelligent Science and Technology, State Key Laboratory of Autonomous Intelligent Unmanned Systems, Tongji University, Shanghai 201804, China,

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ABSTRACT

Audio-visual segmentation (AVS) aims to segment sounding objects at the pixel level by integrating auditory and visual cues. However, existing methods are predominantly developed under the monaural setting and primarily rely on cross-modal semantic correspondence, which limits their ability to distinguish visually similar instances of the same semantic class. In contrast, humans naturally exploit binaural hearing, where interaural differences and direction-dependent acoustic filtering introduced by the head and pinnae provide physically grounded spatial cues for accurate sound source localization. Motivated by this observation, we introduce binaural audio-visual instance segmentation (BiAVIS), a new task that leverages synchronized binaural audio and video frames to segment sounding instances. To advance research on this task, we establish two benchmarks by manually annotating an existing binaural audio-visual dataset and collecting a new real-world dataset, BiAVIS-Bench, in more challenging and diverse scenarios. We further propose a BiAVIS model, which leverages an audio-only sound source localization network to learn spatial and semantic priors for sounding instances from binaural audio. A query-level audio-visual fusion strategy is subsequently introduced to inject these informative priors into the instance segmentation decoder. Extensive experiments conducted on the two proposed benchmarks demonstrate the superior performance of the BiAVIS model over previous monaural AVS methods, especially in resolving instance-level intra-class ambiguity. On the more challenging BiAVIS-Bench, the proposed BiAVIS model outperforms the best-performing monaural baselines by 17.22% in mAP and 7.71% in FSLA, respectively.

1. Introduction

Vision and audition are two fundamental sensory modalities through which humans perceive and understand the world [1]. By integrating visual and auditory cues, people can robustly localize and identify sounding objects in complex scenarios [2]. Inspired by this human perception mechanism, a wide range of audio-visual tasks have been extensively explored in recent years, spanning audio-visual recognition and reasoning [3, 4, 5], audio-visual correspondence [6, 7] and sound source localization (SSL) [8, 9]. More recently, audio-visual research has extended from coarse localization to pixel-level mask prediction of sounding objects, giving rise to audio-visual semantic segmentation (AVSS) [10] and audio-visual instance segmentation (AVIS) [11].

Nevertheless, a fundamental gap persists between human perception and current audio-visual segmentation (AVS) approaches. Humans naturally rely on binaural hearing to localize and distinguish sounding objects [12]. Specifically, the auditory system exploits interaural time differences (ITD) and interaural level differences (ILD) between the two ears, while the head, torso, and pinnae further

impose direction-dependent acoustic filtering on incoming sounds [13]. Together, these cues provide physically grounded spatial information about the relative position of a sound source [14]. In contrast, most existing AVS methods operate on monaural audio and primarily rely on cross-modal semantic correspondence between acoustic and visual representations of sounding objects [15]. For example, CAVP [16] mines informative audio-visual contrastive pairs to better constrain cross-modal embedding learning, while AVSegFormer [17] introduces a transformer-based architecture with audio-conditioned queries to enhance cross-modal interaction. Despite their impressive performance on existing benchmarks, these monaural approaches often struggle in scenarios with multiple candidate instances that belong to the same semantic class, leading to severe intra-class ambiguity. As illustrated in Fig. 1, when a group of people is engaged in the same conversation, different individuals may become the active speaker at different moments. Existing monaural methods can associate audio with the general person category, but often fail to reliably distinguish the active speaker from silent participants. This limitation arises because monaural audio lacks the interaural spatial cues required to distinguish visually similar candidates and reliably resolve instance-level intra-class ambiguity.

To address this limitation, we introduce **Binaural Audio-Visual Instance Segmentation (BiAVIS)**, a new task that leverages synchronized binaural audio and video frames to localize and segment sounding instances within the visual field. Unlike the monaural counterpart, binaural audio

*Corresponding author

✉ wangsaijun@tongji.edu.cn (S. Wang); guanfeng.T@outlook.com (G. Tang); hongbozhao@tongji.edu.cn (H. Zhao); 2453748@tongji.edu.cn (Z. Lei); 2551396@tongji.edu.cn (Y. Zhang); yew@tongji.edu.cn (W. Ye); rui.fan@ieee.org (R. Fan)

ORCID(s):

¹These authors contributed equally to this work.

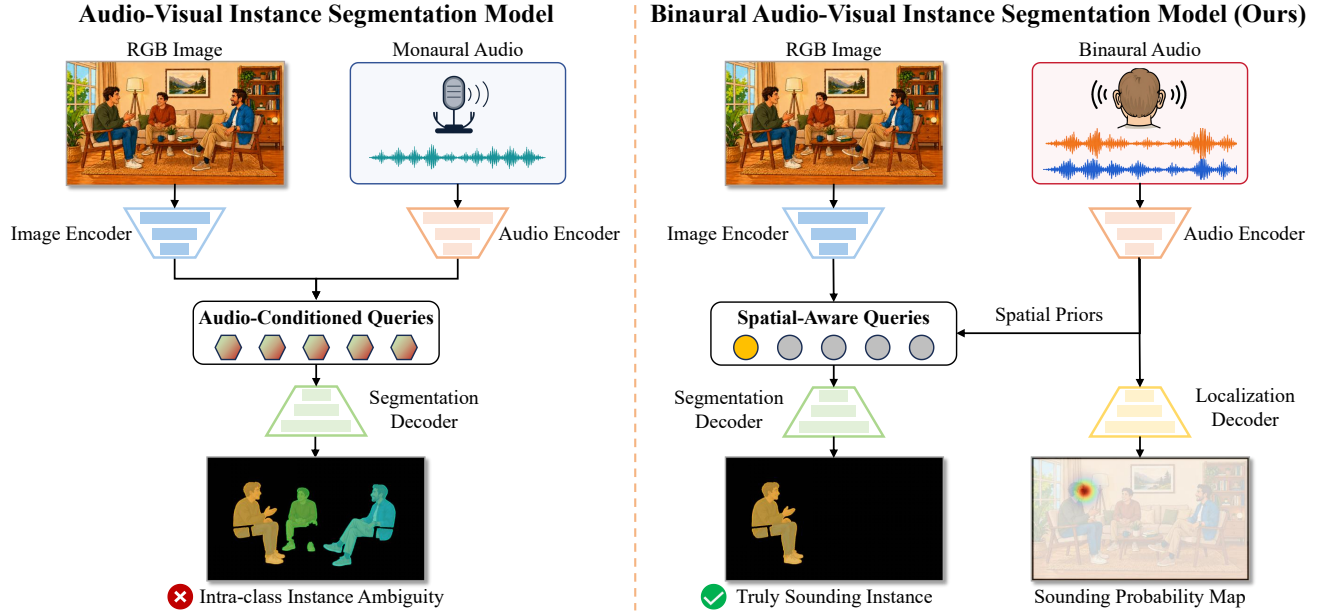


Figure 1: Comparison between AVISM and the proposed BiAVIS. AVISM conditions queries with monaural audio, leading to intra-class instance ambiguity. In contrast, BiAVIS exploits spatial priors, enabling segmentation of the truly sounding instance.

captured by human-like recording devices preserves physically grounded spatial cues associated with sound source direction, which can provide informative spatial priors for audio-visual segmentation. By exploiting these spatial cues, BiAVIS enables more reliable instance-level discrimination and helps resolve intra-class ambiguity in complex scenarios. However, there is currently no publicly available dataset tailored to this task. This is mainly because conventional recording systems typically capture either monaural audio or two-channel audio without human-like head and pinna structures, neither of which can faithfully preserve the binaural spatial cues [18]. To mitigate this data scarcity, we first manually annotate FAIR-Play [19], an existing binaural audio-visual dataset which was originally developed for binaural audio generation, with instance-level masks of sounding objects. Nevertheless, FAIR-Play is collected entirely in a single music room, resulting in extremely limited scene diversity. To further advance research on BiAVIS, we develop a customized data acquisition system comprising a binocular camera and a human-like binaural recording device. Based on this setup, we introduce BiAVIS-Bench, a challenging real-world benchmark covering diverse scenarios, sounding instances, and acoustic environments.

To fully exploit binaural audio, we develop an audio-only SSL network that learns both spatial and semantic priors to guide the subsequent instance segmentation network. Given only binaural audio as input, it is trained to simultaneously predict a dense sounding probability map and class probability maps on the image plane. A class-aware localization loss is further proposed to provide category-aware spatial supervision for the network. This formulation differs from existing binaural SSL methods, which typically rely on visual input or regress low-dimensional source directions, by predicting

dense image-plane localization maps from binaural audio alone.

Building upon the proposed audio-only SSL network, a query-level audio-visual fusion strategy is introduced to inject binaural spatial and semantic priors into the instance segmentation decoder. Specifically, we leverage the intermediate features extracted from binaural audio by the SSL network to condition the object queries in the segmentation decoder. In this way, the object queries are conditioned on audio-derived spatial and semantic cues before interacting with visual features, guiding the decoder to generate audio-aware instance masks and class predictions. Additionally, an audio-visual consistency loss is introduced to align the predicted instance masks with the class-specific sounding probability maps produced by the SSL network. This loss encourages the segmentation decoder to focus on acoustically active regions and suppress visually salient but silent instances. Our contributions are summarized as follows:

- To the best of our knowledge, this is the first work to explore binaural audio-visual instance segmentation, where binaural audio captured with human-like recording devices is used to derive informative spatial and semantic priors for distinguishing sounding instances.
- We establish two benchmarks for BiAVIS by manually annotating an existing binaural audio-visual dataset and constructing a new real-world dataset across more challenging and diverse scenarios using a customized data acquisition system.
- We propose an audio-only sound source localization network to map binaural audio to pixel-wise sounding

and class probability maps on the image plane, thereby learning spatial and semantic priors.

- We introduce a query-level audio-visual fusion strategy to inject learned priors into the instance segmentation decoder, with an audio-visual consistency loss to reduce false activations on silent but visually salient objects.

2. Related Work

2.1. Sound Source Localization

Sound source localization has been a long-standing topic which has received extensive attention in the computer vision and audio communities. Existing methods can be broadly categorized into three groups according to their input modalities. One line of work takes video frames and monaural audio as input, and typically formulates SSL as a cross-modal alignment problem [20, 21, 22]. For example, the study [23] develops an unsupervised two-stream framework that localizes sounding regions through audio-guided visual attention. In indoor environments, multimodal fusion has also been explored to combine complementary visual and acoustic cues for more robust sound source localization [24]. Despite their effectiveness, similar to existing AVS approaches, these monaural SSL methods also suffer from intra-class ambiguity, due to the lack of informative spatial cues for identifying the truly sounding instances. Another line of work incorporates binaural audio into the localization networks to exploit its inherent spatial cues for more accurate localization [9, 25]. For instance, the study [8] selectively fuses the image and binaural audio features through a recurrent neural network to produce a pixel-wise sounding probability map. More recently, audio-visual localization has also been investigated under challenging acoustic conditions; for example, Li et al. [26] jointly address audio denoising and localization under strong unmanned aerial vehicle ego-noise. Although these approaches introduce binaural spatial cues into localization, their training and inference processes still rely heavily on visual input. As a result, they fail to fully exploit the binaural information. The third line of work relies solely on binaural audio for sound source localization, without incorporating visual information. Representative methods typically adopt a transformer-based architecture to extract and fuse features from the input audio, ultimately regressing the sound azimuth with respect to the binaural rig [27]. Nevertheless, such outputs are limited to estimating a single direction in the horizontal plane rather than producing pixel-wise localization maps, making them unsuitable for scenarios with multiple sound sources.

2.2. Audio-Visual Segmentation

Audio-visual segmentation aims to localize and segment sounding objects by integrating auditory and visual cues. The study [10] introduces the audio-visual semantic segmentation task and establishes the AVSBench dataset

by collecting video clips from YouTube. To leverage audio semantics to guide image segmentation, they design a temporal, pixel-wise audio-visual interaction module that explores cross-modal correlations using a standard encoder-decoder architecture. Subsequent studies extend AVS to open-vocabulary settings by bridging vision-language and audio-language foundation models [28]. Recently, inspired by DETR [29] and Mask2Former [30], researchers have increasingly adopted query-based architectures for sounding object segmentation. For example, AVSegFormer [17] incorporates learnable queries that interact with audio features to capture relevant visual semantics, while COMBO [31] explores multi-order bilateral relations across modality, temporal, and pixel levels to enhance the representation of object queries. Nevertheless, these methods remain limited to class-level prediction and cannot distinguish different sounding instances. To address this limitation, AVISM [32] further extends the task to instance-level sounding object segmentation and tracking across video frames. More recently, ACVIS [33] proposes a sound-aware ordinal counting loss that explicitly supervises the number of sounding instances via ordinal regression to prevent visual-only convergence during training. In this paper, we primarily focus on audio-visual instance segmentation by explicitly leveraging binaural audio to provide spatial and semantic priors of sounding instances.

3. Methodology

3.1. Problem Definition

Fig. 2 provides an overview of the proposed framework. We introduce binaural audio-visual instance segmentation (BiAVIS), a new task that extends conventional AVIS from monaural audio to the binaural setting. Specifically, given an RGB image $I \in \mathbb{R}^{H \times W \times 3}$ and its temporally synchronized left and right channel audio $\mathbf{a}_l, \mathbf{a}_r$, where H and W denote the image height and width, BiAVIS aims to identify and segment all visible sounding instances. Let $N \geq 0$ denote the number of ground-truth sounding instances in the image, with the annotation set denoted by $\{(\mathbf{M}_i, c_i)\}_{i=1}^N$, where $\mathbf{M}_i$ denotes the binary mask of the i -th instance and c_i denotes its semantic category. When $N = 0$, the annotation set is empty, corresponding to a no-source sample. In addition, each ground-truth instance is associated with an image-plane sound source point $\mathbf{p}_i = (x_i, y_i)$ for localization supervision. For human speakers, $\mathbf{p}_i$ approximates the mouth location and serves as a proxy for the speech source location [34], while for other categories, it is set to the centroid of the instance mask. BiAVIS outputs a set of predicted sounding instances $\{(\hat{\mathbf{M}}_i, \hat{c}_i)\}_{i=1}^{\hat{N}}$, where $\hat{N} \geq 0$ denotes the number of predicted sounding instances, $\hat{\mathbf{M}}_i$ denotes the i -th predicted mask and $\hat{c}_i$ denotes its corresponding class. When $\hat{N} = 0$, the predicted set is empty, indicating that no sounding instance is detected.

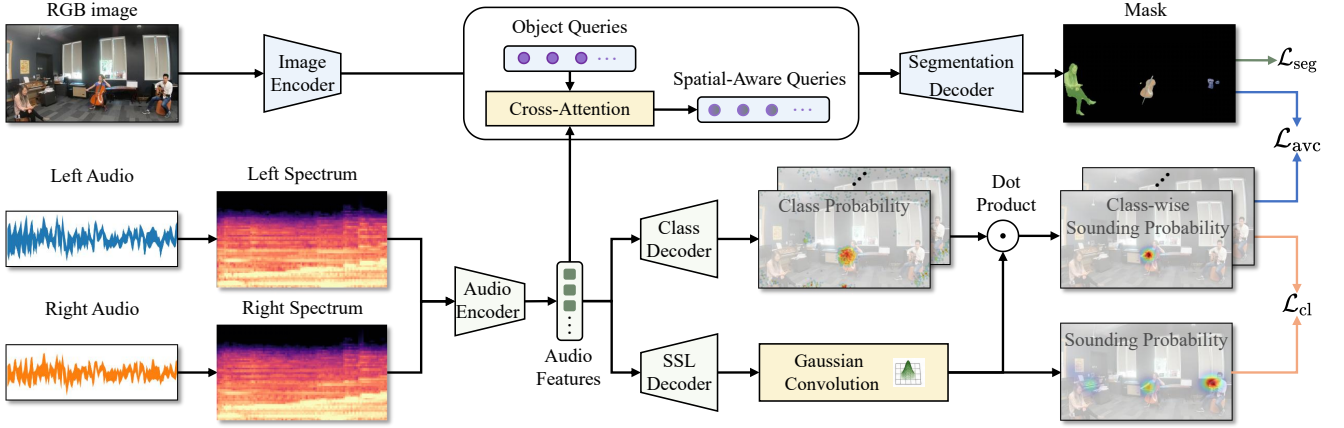


Figure 2: Overview of the proposed BiAVIS framework. The Audio Encoder denotes the complete binaural feature extraction pipeline, consisting of two channel-specific spectrogram encoders followed by a binaural-difference encoder. The audio-only SSL branch predicts sounding probability and class probability maps, while the visual branch follows a Mask2Former-style segmentation decoder. Binaural priors are injected via query-level cross-attention, where object queries are conditioned with binaural features to form audio-aware queries for segmentation prediction. The model is trained with segmentation, class-wise localization, and audio-visual consistency losses.

3.2. Binaural Sound Source Localization

To explicitly extract spatial priors from binaural audio, we design an audio-only sound source localization network that projects binaural spatial cues onto the image plane as sounding probability maps. This audio-to-image projection is enabled by the fixed relative pose between the camera and the binaural recording rig, which makes this projection consistent and learnable.

The binaural audio is first converted into log-mel spectrograms using the short-time Fourier transform, a mel-filterbank, and logarithmic compression. Our audio encoder follows the MambaVision variant of BAST provided in the official BAST-Mamba implementation [27], with the original localization prediction head removed. Each spectrogram is partitioned into non-overlapping time-frequency patches, which are flattened and linearly projected into D -dimensional embeddings. The embeddings are arranged in a fixed time-frequency order, preserving the local structure of the original binaural spectrograms. They are then fed into two architecturally identical but independently parameterized channel-specific spectrogram encoders, producing features F_l and F_r . Motivated by the fact that human sound localization relies on interaural differences, such as ITD and ILD, we feed these features into a binaural-difference encoder to extract spatial and semantic features. The resulting binaural features are denoted by F_b and passed to the subsequent segmentation network.

To supervise the localization network, we construct image-plane supervision from annotated source locations and class labels. A 2D Gaussian response with standard deviation σ is centered at each annotated sound source location, and the responses are summed over the image to obtain the ground-truth map G at a resolution of $\frac{H}{8} \times \frac{W}{8}$. The map is subsequently normalized to obtain $\tilde{G}$, which represents a smoothed sound source probability distribution.

On the prediction side, F_b is pooled into a global embedding and passed to two lightweight feed-forward decoders. The SSL decoder predicts a non-negative sound source activation map $P \in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8}}$, where high responses indicate potential sound source locations on the image plane. Instead of directly predicting a dense Gaussian heatmap, the activation map is encouraged to remain sparse, with most locations having near-zero responses and only a few source-related locations producing high responses. We then convolve the activation map with a Gaussian kernel using the same standard deviation as the target, producing the Gaussian-smoothed response map H and its normalized version $\tilde{H}$. In this way, sparse activations are expanded into local Gaussian responses, yielding probability maps with Gaussian-smoothed response structures that are comparable to the target.

To constrain both response magnitude and spatial distribution shape, we supervise the SSL decoder with a localization loss, which consists of a mean squared error (MSE) loss $\mathcal{L}_{mse}$ and a Jensen-Shannon (JS) divergence [35] loss $\mathcal{L}_{js}$. The MSE loss is computed between the unnormalized maps H and G to constrain pixel-wise response magnitude. In contrast, the JS loss aligns the spatial distribution shape of $\tilde{H}$ with that of $\tilde{G}$ while ignoring the overall response scale. This design restricts the effective prediction space by decoupling source activation prediction from Gaussian spatial smoothing. The decoder only needs to localize potential source centers, while the fixed Gaussian kernel determines the spatial extent of each response and provides a smoother objective than point-only supervision.

Although G provides spatial supervision for sound source locations, it is category-agnostic and cannot specify the category identity of each response peak. Existing AVIS methods commonly encode audio into a global embedding shared across the entire image, which may capture clip-level

sound categories but does not associate them with specific spatial locations. This creates ambiguity in multi-source scenes: a location-only sounding probability map may highlight several source regions without distinguishing which category each peak belongs to. To resolve this ambiguity, we design a class decoder for the SSL network, which predicts class logit maps $\mathbf{Y} \in \mathbb{R}^{C \times \frac{H}{8} \times \frac{W}{8}}$ over C sounding-object categories. It is supervised by a class-wise Gaussian target heatmap $\mathbf{G}^c \in \mathbb{R}^{C \times \frac{H}{8} \times \frac{W}{8}}$, where each annotated sound source point is used as the center of a Gaussian response in its corresponding class channel, following the same process as for $\mathbf{G}$. The class logit maps are converted into class probability maps $\mathbf{P}^c$ by applying a softmax over the class dimension at each spatial location. We supervise the class probability maps with a Gaussian-weighted cross-entropy loss, which can be formulated as follows:

$$\mathcal{L}_c = -\frac{\sum_{c,u,v} \mathbf{G}^c(c, u, v) \log \mathbf{P}^c(c, u, v)}{\sum_{c,u,v} \mathbf{G}^c(c, u, v)}. \quad (1)$$

Here, (u, v) denotes the spatial location on the heatmap grid. For samples without valid sound source annotations, $\mathbf{G}$ and $\mathbf{G}^c$ are set to zero, since they are intensity targets. Accordingly, $\mathcal{L}_c$ is set to zero. For the normalized target, $\tilde{\mathbf{G}}$ is a probability target and is set to a uniform distribution over the heatmap grid, so that it encodes no spatial preference.

The overall class-aware localization loss is defined as:

$$\mathcal{L}_{cl} = \lambda_{js} \mathcal{L}_{js} + \lambda_{mse} \mathcal{L}_{mse} + \lambda_c \mathcal{L}_c, \quad (2)$$

where λ_{js} , λ_{mse} , and λ_c are weighting coefficients that balance the contributions of the JS divergence loss, the MSE loss, and the classification loss, respectively.

3.3. Binaural-Aware Audio-Visual Feature Fusion

For the visual branch, we follow the Mask2Former [30] architecture. Given an input image, a visual backbone and a pixel decoder are used to extract multi-scale visual features. A transformer decoder then takes a fixed number of learnable object queries as input and decodes them into instance masks and class predictions, supervised by the segmentation loss $\mathcal{L}_{seg}$ after bipartite matching with ground-truth sounding instances.

To inject binaural priors into instance segmentation, we adopt a query-level audio-visual feature fusion mechanism. Instead of directly feeding the predicted heatmaps into the visual branch, we project the binaural features $\mathbf{F}_b$ into the decoder feature space and use them as keys and values for cross-attention with the object queries, enabling each query to aggregate audio-derived spatial and semantic cues before visual decoding. To further align the segmentation results with the audio priors, we introduce an audio-visual consistency loss. We define the class-wise sounding probability maps $\mathbf{H}_c$ as the element-wise product between the normalized sounding probability map $\tilde{\mathbf{H}}$ and the c -th class probability map $\mathbf{P}^c$. This map represents the sounding probability that each location belongs to a sounding source of class c . Let $\mathbf{M}_q$ denote the mask probability of the q -th

query. For any query q and class c , we define the coverage score to measure how much of the class-wise sounding probability map is covered by the q -th predicted mask:

$$S(\mathbf{M}_q, \mathbf{H}_c) = \frac{\sum_{u,v} \mathbf{M}_q(u, v) \mathbf{H}_c(u, v)}{\sum_{u,v} \mathbf{M}_q(u, v)}. \quad (3)$$

Let $\mathcal{Q}^+$ denote the set of queries whose masks are matched to ground-truth instances and let $\mathcal{Q}^-$ denote the unmatched set. For a matched query $q \in \mathcal{Q}^+$ assigned to a ground-truth instance of class c_q , we encourage its mask to cover the corresponding class-wise sounding probability map. For an unmatched query $q \in \mathcal{Q}^-$, we suppress its overlap with all class heatmaps by penalizing the maximum coverage score over classes, reducing dataset bias and improving instance selection in multi-source scenes. Therefore, the audio-visual consistency loss is formulated as follows:

$$\mathcal{L}_{avc} = \frac{1}{|\mathcal{Q}^+| + |\mathcal{Q}^-|} \left(-\sum_{q \in \mathcal{Q}^+} \log(S(\mathbf{M}_q, \mathbf{H}_{c_q}) + \epsilon) + \sum_{q \in \mathcal{Q}^-} \max_c S(\mathbf{M}_q, \mathbf{H}_c) \right). \quad (4)$$

Finally, the complete framework is trained end-to-end using the following multi-task objective:

$$\mathcal{L} = \mathcal{L}_{seg} + \lambda_{cl} \mathcal{L}_{cl} + \lambda_{avc} \mathcal{L}_{avc}, \quad (5)$$

where λ_{cl} and λ_{avc} balance the class-aware localization supervision and the audio-visual consistency loss, respectively. This design jointly optimizes instance-level segmentation, binaural SSL, and cross-modal consistency.

4. Experiments

4.1. Datasets

We evaluate BiAVISM on two datasets: FAIR-Play [19] and a newly collected binaural audio-visual dataset, BiAVIS-Bench. FAIR-Play is a public dataset that pairs videos with binaural audio recorded using a fixed custom rig. However, it does not provide the sounding-object annotations required for BiAVIS. To this end, we annotated FAIR-Play with pixel-wise sounding instance masks, semantic labels, and sound source points following the protocol described below. As shown in Fig. 3, the annotated FAIR-Play dataset comprises 1,364 videos and 13,304 clips, covering 8 object categories and 19,164 annotated sounding instances.

Nevertheless, FAIR-Play is collected entirely in a single music room, resulting in extremely limited scene diversity. To better evaluate BiAVISM in more challenging and diverse scenarios, we built a customized data acquisition system and collected a new large-scale benchmark. The system consists of a ZED 2i binocular camera and a Mic.x Binaural I microphone, rigidly mounted on a 3D-printed rig that approximates the relative geometry of human eyes and ears. The camera records videos at 30 fps with a resolution of $1,280 \times 720$ pixels, while the binaural microphone captures

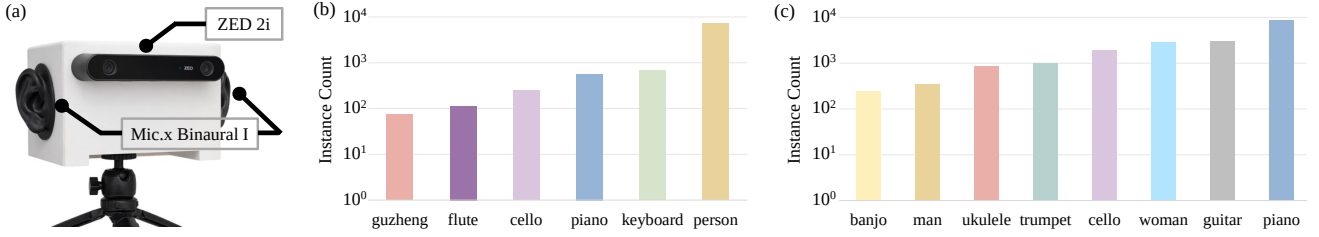


Figure 3: Dataset collection setup and category statistics. (a) Our binaural audio-visual recording system, consisting of a ZED 2i binocular camera and a Mic.x Binaural I microphone mounted on a custom rig. (b) Instance counts per category in BiAVIS-Bench. (c) Instance counts per category in FAIR-Play.

two-channel audio at 48 kHz and 32-bit depth. Audio and video streams are soft-synchronized via a shared ROS2 trigger and temporally aligned using timestamps, with a synchronization error bounded by $10.4 \mu s$. Compared with the FAIR-Play dataset, the proposed BiAVIS-Bench covers a more diverse range of scenes, including indoor, outdoor, and semi-open hall environments. Additionally, it places particular emphasis on scenarios involving multiple instances of the same category. The collected BiAVIS-Bench contains 870 videos and 8,052 clips, covering 6 object categories and 8,656 annotated sounding instances.

For both datasets, videos are divided into one-second clips, and only the center frame of each clip is used for the BiAVIS task. Each sounding instance is annotated with a pixel-level mask, a semantic category label, and a sound source point. Each clip contains up to four simultaneously sounding instances, covering no-source, single-source, and multi-source cases. The training and test splits are constructed at the video level with a ratio of 7:3, ensuring balanced distributions of object categories and instance counts.

We develop a segmentation annotation interface based on X-AnyLabeling. Before large-scale annotation, we evaluate several promptable segmentation models, including SAM, HQ-SAM, MobileSAM, EfficientSAM, and Grounded SAM 2, and select HQ-SAM because it provides the best mask quality. For each one-second clip, annotators first listen to the corresponding audio and identify the visible sounding instances. They then provide click prompts to initialize the instance masks and manually correct inaccurate boundaries or missing regions when necessary. After the initial annotation stage, all annotations underwent multiple rounds of cross-review and correction. The sound source points are generated from the finalized instance masks rather than independently annotated from scratch. For non-human objects, we use the centroid of the instance mask as the source-point proxy. For human speakers, we estimate the approximate mouth location from human body proportions and constrain the resulting point to lie inside the annotated person mask. The generated source points are subsequently subjected to multiple rounds of random-sampling inspection and manual correction.

4.2. Ethics and Privacy

For BiAVIS-Bench, all recorded participants were adults and signed informed consent forms before data collection.

During recruitment and before recording, participants were informed of the recording setup, the types of data collected, the overall research objective of localizing and segmenting sounding instances, the intended research use, and the potential public release of the dataset. No minors or non-consenting bystanders were included, and participants could decline to participate or stop the recording at any time. All recordings were collected in controlled sessions with the participants' consent. The collected data include RGB frames and synchronized binaural audio, which may contain identifiable facial features and voices, but no names or other personally identifying metadata were recorded. Before public release, potentially identifying information will be redacted or anonymized: facial regions will be blurred, human voices will be anonymized through voice conversion, and timestamps and file metadata will be removed to reduce re-identification risk. The dataset does not contain identity-level labels or persistent personal identifiers, and is not designed for identity recognition, biometric profiling, or linking identities across scenes. Accordingly, the proposed model is intended only for localizing and segmenting sounding instances, rather than tracking or recognizing human subjects. We will also specify the dataset license and access terms that prohibit redistribution, re-identification, unauthorized tracking, biometric inference, surveillance use, and any use outside the approved research scope. Users will be required to agree to these terms before accessing the dataset.

4.3. Experimental Setup

Following AVIS [32], we report mean average precision (mAP) and frame-level sound localization accuracy (FSLA) on both FAIR-Play and BiAVIS-Bench to evaluate the segmentation and localization performance of our method against representative approaches. Specifically, mAP evaluates the overall instance-level segmentation quality of sounding objects, while FSLA measures the proportion of frames in which sounding objects are correctly predicted in terms of category and mask overlap. FSLA further includes three sub-metrics: FSLAn, FSLAs, and FSLAm, which evaluate performance in the no-source, single-source, and multi-source cases, respectively. Higher values indicate better performance for all these metrics. The methods compared span three related tasks: video instance segmentation (VIS) [30, 36, 37], audio-visual semantic segmentation (AVSS) [31, 17, 10], and audio-visual instance

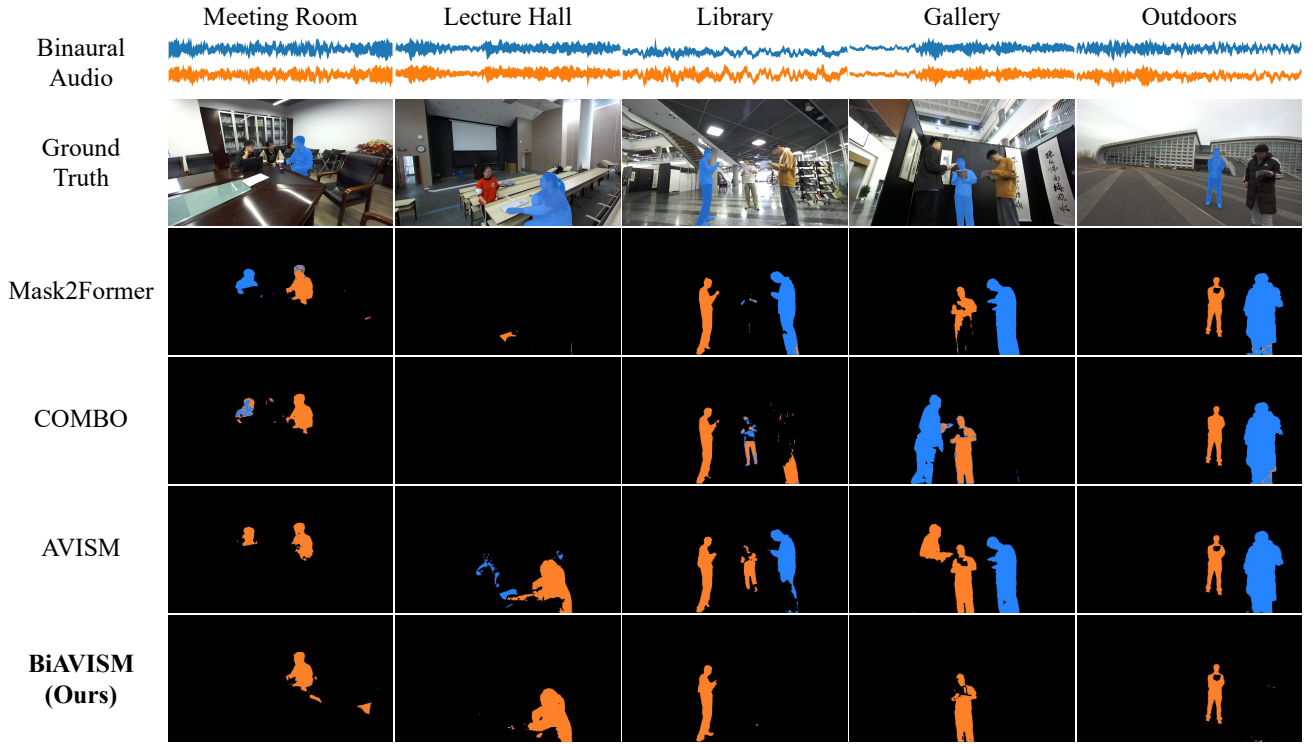


Figure 4: Qualitative comparison of sounding-object segmentation results across diverse scenes.

segmentation (AVIS) [32, 33]. Since VIS methods do not take audio as input, they serve as vision-only instance segmentation baselines. For AVSS methods based on query-based mask prediction, we use their query-level mask outputs to obtain instance-level predictions without changing the model architecture, loss functions, or training procedure. For all methods originally designed for monaural audio, we convert each binaural recording into a single-channel waveform by averaging the synchronized left and right channels. The resulting waveform is then processed using the original audio preprocessing pipeline and model architecture of each baseline. For a fair comparison, all methods in Table 1 use the same ImageNet-pretrained ResNet-50 [38] as the visual backbone, and are trained with the same batch size of 8 and a maximum training budget of 10 epochs. Both the AVSS and AVIS methods adopt Mask2Former-style query-based Transformer decoders, with the primary differences lying in how audio features are extracted, integrated with visual features, and used to guide mask prediction. Training beyond 10 epochs yields no further gains, indicating that all methods have converged under this setting. To evaluate the proposed binaural sound source localization network, we further compare it with representative SSL methods [9, 39, 27] using the evaluation metrics adopted in [8]. The metrics include Pearson’s correlation coefficient (CC), Similarity Metric (SIM), and Earth Mover’s Distance (EMD). Here, CC measures the linear correlation between the predicted and ground-truth heatmaps, SIM measures the similarity between two normalized distributions, and EMD evaluates

the spatial transport cost between them. Higher values are better for CC and SIM, while lower values are better for EMD.

For implementation, we adopt the MambaVision variant of BAST [27] as the audio encoder. The loss weights are set to $\lambda_{js} = 1.0$, $\lambda_{mse} = 1.0$, $\lambda_c = 0.3$, $\lambda_{cl} = 1.2$, and $\lambda_{ave} = 0.5$. All models are optimized using AdamW [40] on a single NVIDIA GeForce RTX 4090 GPU. During training, input images are resized to 640×360 pixels, while evaluation is conducted at the original resolution. For efficiency, single-frame inference at $1,280 \times 720$ pixels has a median latency of about 145 ms and a peak PyTorch memory allocation of about 2.6 GiB over 30 timed runs. Training on FAIR-Play and BiAVIS-Bench takes approximately 3 and 2 hours, respectively.

4.4. Comparison with Methods from Related Tasks

The qualitative comparisons across diverse scenes are shown in Fig. 4, while the quantitative results are reported in Tables 1 and 2. As shown in Fig. 4, competing methods often suffer from intra-class ambiguity, mistakenly segmenting visually salient but non-sounding instances from the same semantic category. In contrast, BiAVISM produces masks that are more consistent with the annotated sounding instances across indoor and outdoor scenarios, suggesting that binaural spatial cues help suppress false activations on visually salient but silent instances. As presented in Table 1, BiAVISM achieves the best overall mAP and FSLA on both

Table 1

Quantitative comparison of different models from related tasks on the FAIR-Play and BiAVIS-Bench test sets.

Task	Networks	Audio	FAIR-Play					BiAVIS-Bench				
			mAP	FSLA	FSLAn	FSLAs	FSLAm	mAP	FSLA	FSLAn	FSLAs	FSLAm
VIS	Mask2Former [30] (CVPR'22)	✗	35.71	60.56	3.52	72.38	44.23	22.32	40.22	10.58	45.07	16.74
	MP-Former [36] (CVPR'23)	✗	26.43	54.17	0.00	73.98	24.40	18.89	19.82	2.88	21.95	10.89
	PolarNeXt [37] (CVPR'25)	✗	30.41	36.55	0.00	51.12	14.45	15.72	28.61	2.88	33.32	4.71
AVSS	COMBO [31] (CVPR'24)	✓	33.27	56.27	17.65	63.86	45.91	19.00	38.49	9.62	43.46	13.88
	AVSegFormer [17] (AAAI'24)	✓	27.36	44.88	76.47	59.75	18.36	15.47	28.61	4.81	32.30	11.15
	AVS-Semantic [10] (IJCV'25)	✓	34.04	53.66	15.29	65.11	36.86	20.14	36.57	7.69	41.51	12.20
AVIS	AVISM [32] (CVPR'25)	✓	33.68	56.94	56.47	66.17	41.63	19.69	40.23	5.77	45.13	18.08
	ACVIS [33] (ICASSP'26)	✓	32.75	60.68	4.71	70.18	48.14	18.35	41.12	5.77	46.04	19.26
	BiAVISM (Ours)	✓	37.63	62.90	89.41	68.63	51.85	23.08	44.29	34.62	48.13	20.69

Table 2

Quantitative comparison of different sound source localization networks on the FAIR-Play and BiAVIS-Bench test sets.

Networks	Image	Monaural	Binaural	FAIR-Play			BiAVIS-Bench		
				CC ↑	SIM ↑	EMD ↓	CC ↑	SIM ↑	EMD ↓
L2BNet [9] (ICCV'21)	✓		✓	0.31	0.14	5.84	0.31	0.22	5.00
Senocak et al. [39] (ICCV'23)	✓	✓		0.16	0.12	5.42	0.13	0.17	5.17
BAST [27] (Neurocomputing'25)			✓	0.17	0.10	6.40	0.09	0.11	6.40
BiAVISM (Ours)			✓	0.38	0.33	3.47	0.37	0.33	2.87

datasets. These results demonstrate that binaural cues help the model distinguish truly sounding instances rather than relying only on visual saliency. Although our model does not achieve the highest FSLAs on FAIR-Play, this is likely due to the dataset's strong single-source bias. When only one object is sounding, it is often also the most visually salient object, allowing visual-only or weakly audio-conditioned baselines to achieve high single-source scores. However, these methods perform much worse in no-source cases, suggesting they are more prone to false positives for visually salient but non-sounding objects. Compared with FAIR-Play, BiAVIS-Bench intentionally includes more challenging scenes with multiple salient objects, especially same-category instances. The advantage of BiAVISM, therefore, becomes more evident on this benchmark, where the model outperforms the best-performing monaural baselines by 17.22% in mAP and 7.71% in FSLA. Overall, these results suggest that extending AVIS from monaural audio to binaural audio improves the model's spatial awareness of sounding instances, and that BiAVIS-Bench serves as a more challenging benchmark for evaluating this ability.

Table 2 further validates the effectiveness of the proposed binaural sound source localization network. Despite using no image input, our model outperforms multimodal localization methods that take images as additional input. On BiAVIS-Bench, it yields relative improvements of 19.35% in

CC and 50.00% in SIM, together with a 42.60% reduction in EMD.

Compared with the binaural audio-only baseline BAST [27], the improvement is even larger, especially on BiAVIS-Bench. These results suggest that visual input alone does not guarantee better localization in this setting, possibly because localization models can become biased toward visually salient regions. In contrast, the proposed image-plane Gaussian supervision enables the SSL network to learn more reliable binaural spatial priors. As shown in Fig. 5, BiAVISM localizes the annotated sound source regions more accurately than existing methods do, which further supports the effectiveness of the proposed audio-only sound source localization network.

4.5. Ablation Studies

4.5.1. Effect of Localization and Consistency Losses

We analyze the effect of localization supervision in Table 3 and the effect of class and audio-visual consistency losses in Table 4. For localization, directly supervising the localization map without Gaussian rendering yields poor alignment with the target distribution, as evidenced by near-zero CC and SIM scores. This shows that directly predicting a dense heatmap is difficult, since the model needs to learn both the sound source activation locations and the spatial extent of each response. By introducing Gaussian rendering, the model only needs to predict sparse source activations,

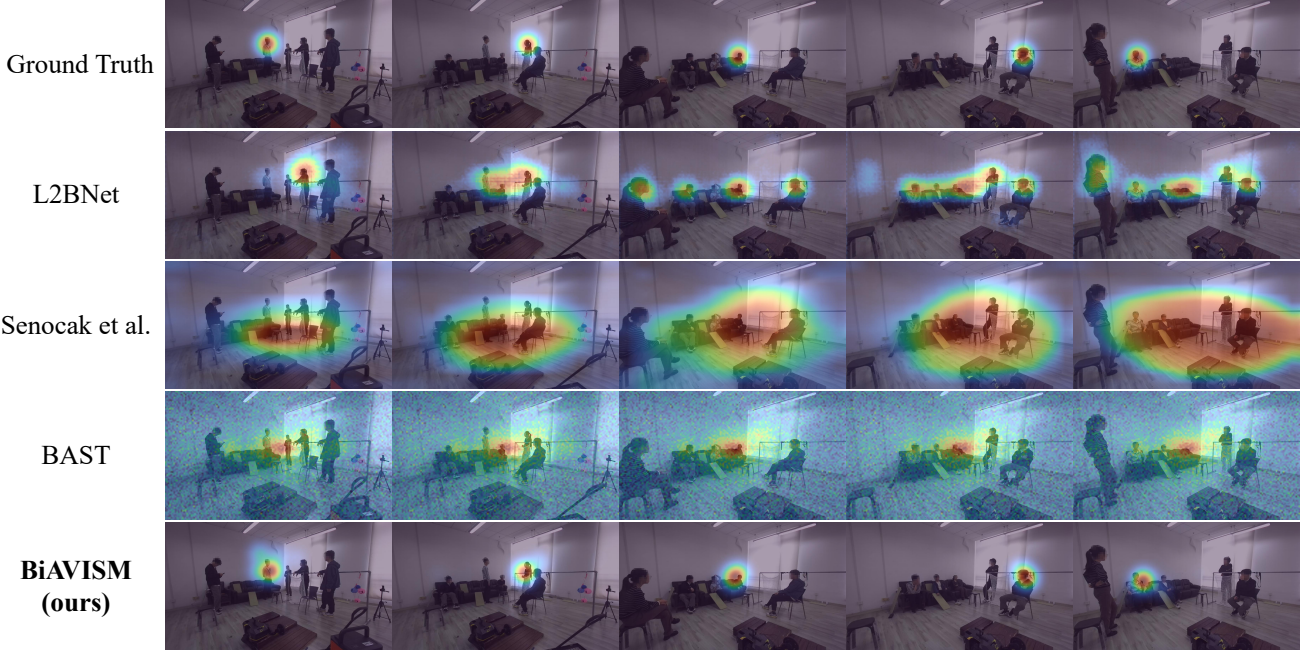


Figure 5: Qualitative comparison of sound source localization methods.

Table 3

Ablation study on the proposed localization loss conducted on the BiAVIS-Bench dataset.

Gaussian	$\mathcal{L}_{mse}$	$\mathcal{L}_{js}$	CC $\uparrow$	SIM $\uparrow$	EMD $\downarrow$
$\times$	$\checkmark$	$\times$	0.01	0.02	5.35
$\times$	$\checkmark$	$\checkmark$	0.06	0.05	4.20
$\checkmark$	$\checkmark$	$\times$	0.31	0.24	3.41
$\checkmark$	$\checkmark$	$\checkmark$	0.37	0.33	2.87

Table 4

Ablation study on the proposed class and audio-visual consistency loss conducted on the BiAVIS-Bench dataset.

$\mathcal{L}_c$	$\mathcal{L}_{avc}$	mAP $\uparrow$	FSLA $\uparrow$	FSLAn $\uparrow$	FSLAs $\uparrow$	FSLAm $\uparrow$
$\times$	$\times$	21.65	30.90	0.00	34.09	19.55
$\times$	$\checkmark$	13.12	33.58	4.81	37.76	14.51
$\checkmark$	$\times$	20.81	32.73	3.85	36.46	16.87
$\checkmark$	$\checkmark$	23.08	44.29	34.62	48.13	20.69

while the fixed Gaussian kernel determines the local spatial extent. The best performance is achieved when Gaussian rendering, $\mathcal{L}_{mse}$, and $\mathcal{L}_{js}$ are used together. This verifies that $\mathcal{L}_{mse}$ and $\mathcal{L}_{js}$ provide complementary supervision, with the former constraining the response magnitude and the latter aligning the normalized spatial distribution.

For class and audio-visual consistency losses, using either loss alone improves FSLA only moderately, with improvements of up to 8.67% with $\mathcal{L}_{avc}$ and 5.92% with $\mathcal{L}_c$. Moreover, $\mathcal{L}_{avc}$ alone even reduces mAP, indicating that the localization map is category-agnostic and cannot reliably

indicate which sound category each response belongs to. When $\mathcal{L}_c$ and $\mathcal{L}_{avc}$ are used together, the model achieves the best overall performance, improving mAP by 6.61% and FSLA by 43.33%. These results validate jointly learning spatial localization and audio category prediction.

4.5.2. Contribution of Binaural Cues

To quantify the contribution of binaural cues, we compare the original binaural input with three monauralized variants in Table 5. The avg setting averages the two channels and duplicates the averaged signal to both channels, while L/L and R/R duplicate the left and right channel respectively, across both input channels; L/R denotes the original binaural input. All variants are trained with the same architecture and training protocol, with only the audio input construction changed. Duplicating or averaging channels removes interaural differences while preserving monaural spectral and semantic information. Compared with the strongest monauralized variant, the original binaural input improves FSLA and FSLAm by 7.69 and 12.83 points on FAIR-Play, and by 1.22 and 1.86 points on BiAVIS-Bench, respectively. The improvement is most pronounced on FSLAm, which evaluates multi-source scenarios that require instance-level spatial discrimination. These results indicate that the performance gain stems from exploiting interaural spatial cues, rather than merely from additional audio information or model capacity.

4.5.3. Audio Encoder Backbone

Our audio encoder follows BAST-Mamba [27], which provides variants based on Vanilla Transformer, Swin Transformer, and MambaVision. We adopt the MambaVision variant because it achieves the best performance in both

Table 5

Ablation study on binaural versus monauralized audio inputs.

Dataset	Audio	mAP $\uparrow$	FSLA $\uparrow$	FSLAn $\uparrow$	FSLAs $\uparrow$	FSLAm $\uparrow$
FAIR-Play	avg	35.14	54.90	88.24	63.29	39.01
	L/L	33.80	54.96	84.71	63.53	39.02
	R/R	37.27	55.21	87.06	64.46	38.02
	L/R	37.63	62.90	89.41	68.63	51.85
BiAVIS-Bench	avg	22.95	43.07	32.69	47.74	13.92
	L/L	21.59	42.43	33.65	47.81	17.84
	R/R	22.91	42.21	33.65	45.96	18.83
	L/R	23.08	44.29	34.62	48.13	20.69

Table 6

Ablation study on audio encoder backbones on the FAIR-Play and BiAVIS-Bench datasets.

Dataset	Implementation	mAP $\uparrow$	FSLA $\uparrow$	FSLAn $\uparrow$	FSLAs $\uparrow$	FSLAm $\uparrow$	CC $\uparrow$	SIM $\uparrow$	EMD $\downarrow$
FAIR-Play	Vanilla Transformer	34.53	57.52	78.82	65.72	42.68	0.34	0.30	3.80
	Swin Transformer	34.81	55.73	70.59	64.88	39.67	0.35	0.30	3.78
	MambaVision	37.63	62.90	89.41	68.63	51.85	0.38	0.33	3.47
BiAVIS-Bench	Vanilla Transformer	18.18	40.97	20.19	45.95	13.35	0.31	0.29	3.15
	Swin Transformer	23.00	41.70	21.15	47.17	10.58	0.31	0.29	3.19
	MambaVision	23.08	44.29	34.62	48.13	20.69	0.37	0.33	2.87

the original BAST experiments and our backbone ablation. Since the encoder processes a two-dimensional time-frequency spectrogram, the official implementation uses MambaVision blocks for this variant. As shown in Table 6, MambaVision consistently outperforms the Vanilla and Swin Transformer variants on both FAIR-Play and BiAVIS-Bench in terms of segmentation metrics (mAP and FSLA) and localization metrics (CC, SIM, and EMD), which justifies our encoder choice.

4.5.4. Audio-Visual Feature Fusion

We further evaluate three audio-visual feature fusion strategies in Table 7. Following the AVIS [32] fusion strategy, global audio embedding addition injects coarse clip-level audio information into the segmentation decoder. However, applying the same global audio embedding to all object queries provides identical information and lacks query-specific spatial priors, leading to limited performance in multi-source scenes.

Another strategy is heatmap-peak conditioning, which is a natural alternative based on our sound source localization network’s output. Specifically, we apply non-maximum suppression to each class heatmap, select the global top- k peaks according to their response values, and feed the peak locations and values into an MLP to condition the object queries. This strategy uses the predicted localization results more explicitly, but it relies on the final heatmap peaks and discards audio semantic cues. As shown in Table 7, compared with global audio embedding addition, heatmap-peak conditioning improves FSLAm by 7.63%, suggesting

that explicit peak locations can provide some benefits for multi-source cases. However, it decreases FSLAs and FSLAn by 16.06% and 50.00%, respectively, indicating that relying only on discrete heatmap peaks is unstable and may hurt single-source and no-source prediction.

The third strategy is our query-level audio conditioning, in which the binaural features F_b serve as audio-side context and are injected into object queries via cross-attention. By allowing each query to adaptively aggregate audio-side spatial and class cues before interacting with visual features, the proposed fusion strategy better guides the decoder toward sounding instances. As shown in Table 7, query-level audio conditioning achieves the best performance across all metrics, improving mAP and FSLA by 18.12% and 12.96% over the strongest alternative strategy, respectively. The gains are especially clear on FSLAn, indicating that query-level conditioning helps suppress false activations in no-source cases.

4.6. Limitations

Despite the promising results, the proposed BiAVISM still has two main limitations. First, the current formulation assumes a fixed geometric relationship between the camera and the binaural recording device, since the audio-only sound source localization network learns to project binaural cues onto the image plane. When the camera-microphone configuration or pose changes, the model may therefore require fine-tuning. Second, while BiAVIS-Bench covers a more diverse range of scenes than FAIR-Play, its scale and category coverage remain limited compared with large-scale

Table 7

Ablation study on the audio-visual feature fusion strategies.

Fusion strategy	mAP $\uparrow$	FSLA $\uparrow$	FSLAn $\uparrow$	FSLAs $\uparrow$	FSLAm $\uparrow$
Global audio embedding addition	19.49	39.21	11.54	43.53	18.74
Heatmap-peak conditioning	19.54	33.27	5.77	36.54	20.17
Query-level audio conditioning	23.08	44.29	34.62	48.13	20.69

image or video segmentation datasets. This may restrict the model's generalization to unseen object categories and more complex acoustic environments.

5. Conclusion and Future Work

This paper studied binaural audio-visual instance segmentation (BiAVIS), which aims to identify and segment visible sounding instances from synchronized RGB frames and human-like binaural audio. Unlike monaural AVS pipelines that mainly exploit cross-modal semantic correspondence, BiAVIS emphasizes physically grounded spatial cues carried by interaural differences and direction-dependent acoustic filtering. To support this setting, we annotated FAIR-Play with instance-level sounding masks and collected BiAVIS-Bench using a binocular camera and a human-like binaural microphone in diverse indoor, outdoor, and semi-open scenes. We further proposed BiAVISM, which first learns image-plane spatial and semantic priors through an audio-only sound source localization branch, and then injects these priors into a query-based instance segmentation decoder via query-level audio-visual fusion.

The work also has clear weaknesses. As discussed in Section 4.6, BiAVISM currently assumes a fixed camera-microphone geometry, so changes in device layout or relative pose may require fine-tuning. In addition, although BiAVIS-Bench is more diverse than FAIR-Play, its scale and category coverage remain limited relative to large-scale vision datasets, which can constrain generalization to rare objects and complex acoustic mixtures. From a deployment perspective, RGB frames and binaural audio may contain identifiable faces and voices; therefore, dataset release and model use must remain consent-aware and privacy-preserving, as outlined in Section 4.2.

Beyond our own experiments, we expect the community to benefit from BiAVIS in several ways. The annotated FAIR-Play split and BiAVIS-Bench provide a starting point for comparing binaural AVS methods under a shared protocol. More broadly, reliable sounding-instance segmentation may support audio-visual scene understanding, assistive perception, and human-centered robotic systems that need to attend to the objects that are actually producing sound, rather than all visually salient candidates.

Several directions remain open for future work. One priority is calibration-aware or calibration-free binaural learning, so that models can transfer across different camera-microphone baselines, head-related filtering characteristics, and recording devices without expensive re-annotation. A

second direction is to enlarge the benchmark with more categories, denser multi-source overlap, and stronger reverberation and noise. Finally, although the present study focuses on frame-level BiAVIS to validate the benefit of binaural spatial cues for resolving instance-level intra-class ambiguity, extending it to temporal instance segmentation remains an important next step. Under realistic binaural sensing conditions, we hope BiAVIS will encourage further work on spatially grounded multimodal recognition.

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